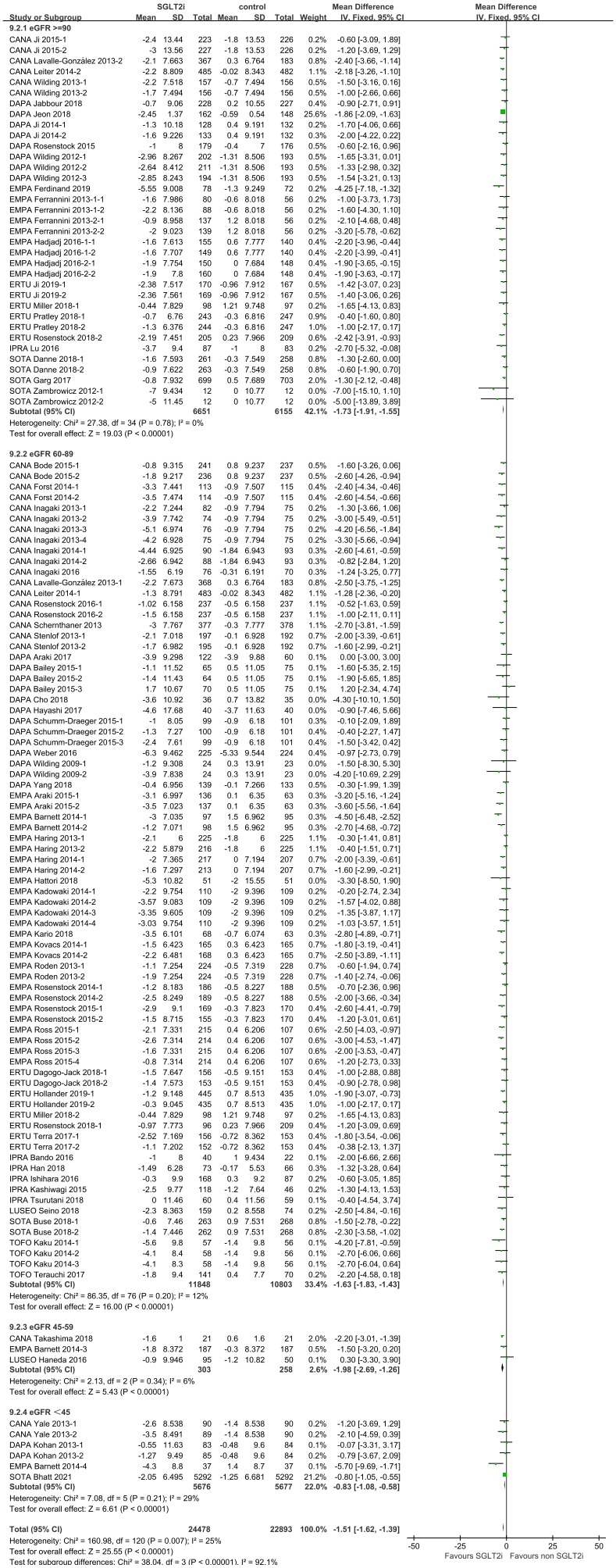

SGLT2i
control
Mean Difference
Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Fixed, 95% CI
IV, Fixed, 95% CI
9.2.1 eGFR >=90
CANA Ji 2015-1 -2.4 13.44 223 -1.8 13.53 226 0.2% -0.60 [-3.09, 1.89]
CANA Ji 2015-2 -3 13.56 227 -1.8 13.53 226 0.2% -1.20 [-3.69, 1.29]
CANA Lavalle-González 2013-2 -2.1 7.663 367 0.3 6.764 183 0.8% -2.40 [-3.66, -1.14]
CANA Leiter 2014-2 -2.2 8.809 485 -0.02 8.343 482 1.1% -2.18 [-3.26, -1.10]
CANA Wilding 2013-1 -2.2 7.518 157 -0.7 7.494 156 0.5% -1.50 [-3.16, 0.16]
CANA Wilding 2013-2 -1.7 7.494 156 -0.7 7.494 156 0.5% -1.00 [-2.66, 0.66]
DAPA Jabbour 2018 -0.7 9.06 228 0.2 10.55 227 0.4% -0.90 [-2.71, 0.91]
DAPA Jeon 2018 -2.45 1.37 162 -0.59 0.54 148 25.6% -1.86 [-2.09, -1.63]
DAPA Ji 2014-1 -1.3 10.18 128 0.4 9.191 132 0.2% -1.70 [-4.06, 0.66]
DAPA Ji 2014-2 -1.6 9.226 133 0.4 9.191 132 0.3% -2.00 [-4.22, 0.22]
DAPA Rosenstock 2015 -1 8 179 -0.4 7 176 0.5% -0.60 [-2.16, 0.96]
DAPA Wilding 2012-1 -2.96 8.267 202 -1.31 8.506 193 0.5% -1.65 [-3.31, 0.01]
DAPA Wilding 2012-2 -2.64 8.412 211 -1.31 8.506 193 0.5% -1.33 [-2.98, 0.32]
DAPA Wilding 2012-3 -2.85 8.243 194 -1.31 8.506 193 0.5% -1.54 [-3.21, 0.13]
EMPA Ferdinand 2019 -5.55 9.008 78 -1.3 9.249 72 0.2% -4.25 [-7.18, -1.32]
EMPA Ferrannini 2013-1-1 -1.6 7.986 80 -0.6 8.018 56 0.2% -1.00 [-3.73, 1.73]
EMPA Ferrannini 2013-1-2 -2.2 8.136 88 -0.6 8.018 56 0.2% -1.60 [-4.30, 1.10]
EMPA Ferrannini 2013-2-1 -0.9 8.958 137 1.2 8.018 56 0.2% -2.10 [-4.68, 0.48]
EMPA Ferrannini 2013-2-2 -2 9.023 139 1.2 8.018 56 0.2% -3.20 [-5.78, -0.62]
EMPA Hadjadj 2016-1-1 -1.6 7.613 155 0.6 7.777 140 0.4% -2.20 [-3.96, -0.44]
EMPA Hadjadj 2016-1-2 -1.6 7.707 149 0.6 7.777 140 0.4% -2.20 [-3.99, -0.41]
EMPA Hadjadj 2016-2-1 -1.9 7.754 150 0 7.684 148 0.4% -1.90 [-3.65, -0.15]
EMPA Hadjadj 2016-2-2 -1.9 7.8 160 0 7.684 148 0.4% -1.90 [-3.63, -0.17]
ERTU Ji 2019-1 -2.38 7.517 170 -0.96 7.912 167 0.5% -1.42 [-3.07, 0.23]
ERTU Ji 2019-2 -2.36 7.561 169 -0.96 7.912 167 0.5% -1.40 [-3.06, 0.26]
ERTU Miller 2018-1 -0.44 7.829 98 1.21 9.748 97 0.2% -1.65 [-4.13, 0.83]
ERTU Pratley 2018-1 -0.7 6.76 243 -0.3 6.816 247 0.9% -0.40 [-1.60, 0.80]
ERTU Pratley 2018-2 -1.3 6.376 244 -0.3 6.816 247 1.0% -1.00 [-2.17, 0.17]
ERTU Rosenstock 2018-2 -2.19 7.451 205 0.23 7.966 209 0.6% -2.42 [-3.91, -0.93]
IPRA Lu 2016 -3.7 9.4 87 -1 8 83 0.2% -2.70 [-5.32, -0.08]
SOTA Danne 2018-1 -1.6 7.593 261 -0.3 7.549 258 0.8% -1.30 [-2.60, 0.00]
SOTA Danne 2018-2 -0.9 7.622 263 -0.3 7.549 258 0.8% -0.60 [-1.90, 0.70]
SOTA Garg 2017 -0.8 7.932 699 0.5 7.689 703 2.0% -1.30 [-2.12, -0.48]
SOTA Zambrowicz 2012-1 -7 9.434 12 0 10.77 12 0.0% -7.00 [-15.10, 1.10]
SOTA Zambrowicz 2012-2 -5 11.45 12 0 10.77 12 0.0% -5.00 [-13.89, 3.89]
Subtotal (95% CI) 6651 6155 42.1% -1.73 [-1.91, -1.55]
Heterogeneity: Chi² = 27.38, df = 34 (P = 0.78); I² = 0%
Test for overall effect: Z = 19.03 (P < 0.00001)
9.2.2 eGFR 60-89
CANA Bode 2015-1 -0.8 9.315 241 0.8 9.237 237 0.5% -1.60 [-3.26, 0.06]
CANA Bode 2015-2 -1.8 9.217 236 0.8 9.237 237 0.5% -2.60 [-4.26, -0.94]
CANA Forst 2014-1 -3.3 7.441 113 -0.9 7.507 115 0.4% -2.40 [-4.34, -0.46]
CANA Forst 2014-2 -3.5 7.474 114 -0.9 7.507 115 0.4% -2.60 [-4.54, -0.66]
CANA Inagaki 2013-1 -2.2 7.244 82 -0.9 7.794 75 0.2% -1.30 [-3.66, 1.06]
CANA Inagaki 2013-2 -3.9 7.742 74 -0.9 7.794 75 0.2% -3.00 [-5.49, -0.51]
CANA Inagaki 2013-3 -5.1 6.974 76 -0.9 7.794 75 0.2% -4.20 [-6.56, -1.84]
CANA Inagaki 2013-4 -4.2 6.928 75 -0.9 7.794 75 0.2% -3.30 [-5.66, -0.94]
CANA Inagaki 2014-1 -4.44 6.925 90 -1.84 6.943 93 0.3% -2.60 [-4.61, -0.59]
CANA Inagaki 2014-2 -2.66 6.942 88 -1.84 6.943 93 0.3% -0.82 [-2.84, 1.20]
CANA Inagaki 2016 -1.55 6.19 76 -0.31 6.191 70 0.3% -1.24 [-3.25, 0.77]
CANA Lavalle-González 2013-1 -2.2 7.673 368 0.3 6.764 183 0.8% -2.50 [-3.75, -1.25]
CANA Leiter 2014-1 -1.3 8.791 483 -0.02 8.343 482 1.1% -1.28 [-2.36, -0.20]
CANA Rosenstock 2016-1 -1.02 6.158 237 -0.5 6.158 237 1.1% -0.52 [-1.63, 0.59]
CANA Rosenstock 2016-2 -1.5 6.158 237 -0.5 6.158 237 1.1% -1.00 [-2.11, 0.11]
CANA Schernthaner 2013 -3 7.767 377 -0.3 7.777 378 1.1% -2.70 [-3.81, -1.59]
CANA Stenlof 2013-1 -2.1 7.018 197 -0.1 6.928 192 0.7% -2.00 [-3.39, -0.61]
CANA Stenlof 2013-2 -1.7 6.982 195 -0.1 6.928 192 0.7% -1.60 [-2.99, -0.21]
DAPA Araki 2017 -3.9 9.298 122 -3.9 9.88 60 0.1% 0.00 [-3.00, 3.00]
DAPA Bailey 2015-1 -1.1 11.52 65 0.5 11.05 75 0.1% -1.60 [-5.35, 2.15]
DAPA Bailey 2015-2 -1.4 11.43 64 0.5 11.05 75 0.1% -1.90 [-5.65, 1.85]
DAPA Bailey 2015-3 1.7 10.67 70 0.5 11.05 75 0.1% 1.20 [-2.34, 4.74]
DAPA Cho 2018 -3.6 10.92 36 0.7 13.82 35 0.0% -4.30 [-10.10, 1.50]
DAPA Hayashi 2017 -4.6 17.68 40 -3.7 11.63 40 0.0% -0.90 [-7.46, 5.66]
DAPA Schumm-Draeger 2015-1 -1 8.05 99 -0.9 6.18 101 0.3% -0.10 [-2.09, 1.89]
DAPA Schumm-Draeger 2015-2 -1.3 7.27 100 -0.9 6.18 101 0.4% -0.40 [-2.27, 1.47]
DAPA Schumm-Draeger 2015-3 -2.4 7.61 99 -0.9 6.18 101 0.4% -1.50 [-3.42, 0.42]
DAPA Weber 2016 -6.3 9.462 225 -5.33 9.544 224 0.4% -0.97 [-2.73, 0.79]
DAPA Wilding 2009-1 -1.2 9.308 24 0.3 13.91 23 0.0% -1.50 [-8.30, 5.30]
DAPA Wilding 2009-2 -3.9 7.838 24 0.3 13.91 23 0.0% -4.20 [-10.69, 2.29]
DAPA Yang 2018 -0.4 6.956 139 -0.1 7.266 133 0.5% -0.30 [-1.99, 1.39]
EMPA Araki 2015-1 -3.1 6.997 136 0.1 6.35 63 0.3% -3.20 [-5.16, -1.24]
EMPA Araki 2015-2 -3.5 7.023 137 0.1 6.35 63 0.3% -3.60 [-5.56, -1.64]
EMPA Barnett 2014-1 -3 7.035 97 1.5 6.962 95 0.3% -4.50 [-6.48, -2.52]
EMPA Barnett 2014-2 -1.2 7.071 98 1.5 6.962 95 0.3% -2.70 [-4.68, -0.72]
EMPA Haring 2013-1 -2.1 6 225 -1.8 6 225 1.1% -0.30 [-1.41, 0.81]
EMPA Haring 2013-2 -2.2 5.879 216 -1.8 6 225 1.1% -0.40 [-1.51, 0.71]
EMPA Haring 2014-1 -2 7.365 217 0 7.194 207 0.7% -2.00 [-3.39, -0.61]
EMPA Haring 2014-2 -1.6 7.297 213 0 7.194 207 0.7% -1.60 [-2.99, -0.21]
EMPA Hattori 2018 -5.3 10.82 51 -2 15.55 51 0.0% -3.30 [-8.50, 1.90]
EMPA Kadowaki 2014-1 -2.2 9.754 110 -2 9.396 109 0.2% -0.20 [-2.74, 2.34]
EMPA Kadowaki 2014-2 -3.57 9.083 109 -2 9.396 109 0.2% -1.57 [-4.02, 0.88]
EMPA Kadowaki 2014-3 -3.35 9.605 109 -2 9.396 109 0.2% -1.35 [-3.87, 1.17]
EMPA Kadowaki 2014-4 -3.03 9.754 110 -2 9.396 109 0.2% -1.03 [-3.57, 1.51]
EMPA Kario 2018 -3.5 6.101 68 -0.7 6.074 63 0.3% -2.80 [-4.89, -0.71]
EMPA Kovacs 2014-1 -1.5 6.423 165 0.3 6.423 165 0.7% -1.80 [-3.19, -0.41]
EMPA Kovacs 2014-2 -2.2 6.481 168 0.3 6.423 165 0.7% -2.50 [-3.89, -1.11]
EMPA Roden 2013-1 -1.1 7.254 224 -0.5 7.319 228 0.7% -0.60 [-1.94, 0.74]
EMPA Roden 2013-2 -1.9 7.254 224 -0.5 7.319 228 0.7% -1.40 [-2.74, -0.06]
EMPA Rosenstock 2014-1 -1.2 8.183 186 -0.5 8.227 188 0.5% -0.70 [-2.36, 0.96]
EMPA Rosenstock 2014-2 -2.5 8.249 189 -0.5 8.227 188 0.5% -2.00 [-3.66, -0.34]
EMPA Rosenstock 2015-1 -2.9 9.1 169 -0.3 7.823 170 0.4% -2.60 [-4.41, -0.79]
EMPA Rosenstock 2015-2 -1.5 8.715 155 -0.3 7.823 170 0.4% -1.20 [-3.01, 0.61]
EMPA Ross 2015-1 -2.1 7.331 215 0.4 6.206 107 0.6% -2.50 [-4.03, -0.97]
EMPA Ross 2015-2 -2.6 7.314 214 0.4 6.206 107 0.6% -3.00 [-4.53, -1.47]
EMPA Ross 2015-3 -1.6 7.331 215 0.4 6.206 107 0.6% -2.00 [-3.53, -0.47]
EMPA Ross 2015-4 -0.8 7.314 214 0.4 6.206 107 0.6% -1.20 [-2.73, 0.33]
ERTU Dagogo-Jack 2018-1 -1.5 7.647 156 -0.5 9.151 153 0.4% -1.00 [-2.88, 0.88]
ERTU Dagogo-Jack 2018-2 -1.4 7.573 153 -0.5 9.151 153 0.4% -0.90 [-2.78, 0.98]
ERTU Hollander 2019-1 -1.2 9.148 445 0.7 8.513 435 1.0% -1.90 [-3.07, -0.73]
ERTU Hollander 2019-2 -0.3 9.045 435 0.7 8.513 435 1.0% -1.00 [-2.17, 0.17]
ERTU Miller 2018-2 -0.44 7.829 98 1.21 9.748 97 0.2% -1.65 [-4.13, 0.83]
ERTU Rosenstock 2018-1 -0.97 7.773 96 0.23 7.966 209 0.4% -1.20 [-3.09, 0.69]
ERTU Terra 2017-1 -2.52 7.169 156 -0.72 8.362 153 0.4% -1.80 [-3.54, -0.06]
ERTU Terra 2017-2 -1.1 7.202 152 -0.72 8.362 153 0.4% -0.38 [-2.13, 1.37]
IPRA Bando 2016 -1 8 40 1 9.434 22 0.1% -2.00 [-6.66, 2.66]
IPRA Han 2018 -1.49 6.28 73 -0.17 5.53 66 0.3% -1.32 [-3.28, 0.64]
IPRA Ishihara 2016 -0.3 9.9 168 0.3 9.2 87 0.2% -0.60 [-3.05, 1.85]
IPRA Kashiwagi 2015 -2.5 9.77 118 -1.2 7.64 46 0.2% -1.30 [-4.13, 1.53]
IPRA Tsurutani 2018 0 11.46 60 0.4 11.56 59 0.1% -0.40 [-4.54, 3.74]
LUSEO Seino 2018 -2.3 8.363 159 0.2 8.558 74 0.2% -2.50 [-4.84, -0.16]
SOTA Buse 2018-1 -0.6 7.46 263 0.9 7.531 268 0.8% -1.50 [-2.78, -0.22]
SOTA Buse 2018-2 -1.4 7.446 262 0.9 7.531 268 0.8% -2.30 [-3.58, -1.02]
TOFO Kaku 2014-1 -5.6 9.8 57 -1.4 9.8 56 0.1% -4.20 [-7.81, -0.59]
TOFO Kaku 2014-2 -4.1 8.4 58 -1.4 9.8 56 0.1% -2.70 [-6.06, 0.66]
TOFO Kaku 2014-3 -4.1 8.3 58 -1.4 9.8 56 0.1% -2.70 [-6.04, 0.64]
TOFO Terauchi 2017 -1.8 9.4 141 0.4 7.7 70 0.2% -2.20 [-4.58, 0.18]
Subtotal (95% CI) 11848 10803 33.4% -1.63 [-1.83, -1.43]
Heterogeneity: Chi² = 86.35, df = 76 (P = 0.20); I² = 12%
Test for overall effect: Z = 16.00 (P < 0.00001)
9.2.3 eGFR 45-59
CANA Takashima 2018 -1.6 1 21 0.6 1.6 21 2.0% -2.20 [-3.01, -1.39]
EMPA Barnett 2014-3 -1.8 8.372 187 -0.3 8.372 187 0.5% -1.50 [-3.20, 0.20]
LUSEO Haneda 2016 -0.9 9.946 95 -1.2 10.82 50 0.1% 0.30 [-3.30, 3.90]
Subtotal (95% CI) 303 258 2.6% -1.98 [-2.69, -1.26]
Heterogeneity: Chi² = 2.13, df = 2 (P = 0.34); I² = 6%
Test for overall effect: Z = 5.43 (P < 0.00001)
9.2.4 eGFR <45
CANA Yale 2013-1 -2.6 8.538 90 -1.4 8.538 90 0.2% -1.20 [-3.69, 1.29]
CANA Yale 2013-2 -3.5 8.491 89 -1.4 8.538 90 0.2% -2.10 [-4.59, 0.39]
DAPA Kohan 2013-1 -0.55 11.63 83 -0.48 9.6 84 0.1% -0.07 [-3.31, 3.17]
DAPA Kohan 2013-2 -1.27 9.49 85 -0.48 9.6 84 0.2% -0.79 [-3.67, 2.09]
EMPA Barnett 2014-4 -4.3 8.8 37 1.4 8.7 37 0.1% -5.70 [-9.69, -1.71]
SOTA Bhatt 2021 -2.05 6.495 5292 -1.25 6.681 5292 21.2% -0.80 [-1.05, -0.55]
Subtotal (95% CI) 5676 5677 22.0% -0.83 [-1.08, -0.58]
Heterogeneity: Chi² = 7.08, df = 5 (P = 0.21); I² = 29%
Test for overall effect: Z = 6.61 (P < 0.00001)
Total (95% CI) 24478 22893 100.0% -1.51 [-1.62, -1.39]
Heterogeneity: Chi² = 160.98, df = 120 (P = 0.007); I² = 25%
Test for overall effect: Z = 25.55 (P < 0.00001)
Test for subgroup differences: Chi² = 38.04, df = 3 (P < 0.00001), I² = 92.1%
-50
-25
0
25
50
Favours SGLT2i
Favours non SGLT2i